Solution:

$$\frac{\frac{1}{1!}+\frac{1}{3}+\frac{1}{5!}+\ldots}{\frac{1}{2!}+\frac{1}{4!}+\frac{1}{6!}+\ldots} = \frac{\frac{e-e^{-1}}{2}}{\left[\frac{e+e^{-1}}{2}-1\right]}$$

$$= \left[\frac{e-e^{-1}}{2}\right] \div \left[\frac{e+e^{-1}-2}{2}\right]$$

$$= \frac{e-e^{-1}}{e+e^{-1}-2}$$

$$= \left[e-\frac{1}{e}\right] \div \left[e+\frac{1}{e}-2\right]$$

$$= \left[\frac{e^2-1}{e}\right] \div \left[\frac{e^2-2e+1}{e}\right]$$

$$= \frac{(e^2-1)}{(e^2-2e+1)}$$

$$= \frac{(e^2-1)}{(e-1)^2}$$

$$= \frac{(e-1)(e+1)}{(e-1)^2}$$

$$= \frac{e+1}{e-1}$$

7.4.3 Logarithmic series:

If $|x| < 1$, the series, $x - \frac{x^2}{2}+\frac{x^3}{3}-\frac{x^4}{4}+\ldots$ converges to $\log(1+x)$

Some important deductions from the above series are

(i) $\log(1+x) = x - \frac{x^2}{2}+\frac{x^3}{3}-\frac{x^4}{4}+\ldots$

(ii) $\log(1-x) = -x - \frac{x^2}{2}-\frac{x^3}{3}-\frac{x^4}{4}+\ldots$

(iii) $\log(1+x) - \log(1-x) = 2\left(x+\frac{x^3}{3}+\frac{x^5}{5}+\ldots\right)$

or $\frac{1}{2}\log\left(\frac{1+x}{1-x}\right) = x+\frac{x^3}{3}+\frac{x^5}{5}+\ldots$

7.5 Introduction to Elementary calculus

Before going to understand the problems on continuous random variables, we need to know some fundamental knowledge about differentiation and integration, which are part of calculus in higher mathematics.

Hence, we introduce some simple concepts, techniques and formulae to calculate problems in statistics, which involve calculus.